Hustle City Product Requirements

Founder build plan and PC to homelab deployment

Version 1.0 • 11 October 2026 • Owner Oghenemaro • Status Proposed implementation baseline

Product decision

Build a Nigerian playable marketplace for adults aged 18 to 35. Players explore a fictional city, manage a simulated shop, meet friends, discover real businesses and artisans, complete paid work, and receive eligible earnings in Nigerian bank accounts. Merchants get a free digital storefront and can upgrade to premium shop, workshop or office templates.

Start on a development PC, deploy the same containerised application to the Dell Proxmox homelab for a private beta, then assess public hosting before enabling live payouts. This document defines the full vision and a smaller first release. Later phases preserve the requested multiplayer, 3D interiors and complex trading features.

Problem and value

Small merchants and artisans need affordable discovery, catalogues and customer enquiries. Young adults want an entertaining social experience with credible economic opportunities. Hustle City connects these needs through one marketplace accessed either by walking through a virtual neighbourhood or using quick search.

First release promise

- A player can explore Junction 101, create an avatar, run a simulated shop, complete daily quests and connect with friends.
- A merchant can publish a free real shop, workshop or office; a customer can find products, request pickup, or request an artisan quote.
- A worker can apply for a real digital task, submit evidence and receive a review. Earnings and payouts are exercised in sandbox before live use.

Success criteria

Private beta planning cohort: 30 to 50 players and 10 merchants in one real Lagos service area. Proposed targets: 20% day 7 player retention, 10 completed marketplace interactions, 3 repeat customers within 30 days and zero unexplained ledger discrepancies. These are hypotheses to validate, not forecasts.

Assumptions to confirm

The exact Lagos service area, merchant categories, founder weekly capacity, development PC specification and payment partner approval remain open. Use electronics retailers and repair artisans as the initial recruiting hypothesis. Home servers provide no redundancy by default.

World and user journeys

Story and neighbourhood

The player arrives in New Lagos to build a business and social life. Junction 101 is a fictional compact neighbourhood with recognisable Nigerian architecture, English and optional Pidgin dialogue. Real merchants retain their actual pickup address, service area and delivery coverage regardless of their virtual location.

Location	Interactive purpose
Market Square	Simulated shop management and real merchant storefronts, product browsing requests.
Work Hub	Digital job listings, applications, submissions and earnings history.
The Hangout	Friends, clubs and basic leaderboards.

Only these three locations introduce gameplay in the first release. Apartments, offices and workshops are reusable space templates within the locations or map scenery; they do not require a separate residential system.

Player onboarding

Create a nickname and avatar, enter the map, receive 2,000 City Coins, visit the three locations, complete one simulated order and claim the introductory quest. Show that City Coins have no cash value. Allow real marketplace browsing without forcing the tutorial or game movement.

Customer and artisan journey

Search generator repair using an actual service location; compare profiles, coverage and transaction reviews; send a description and optional photo; receive an itemised quote and appointment; accept the quote; record completion; pay using the approved flow; review the provider. Additional charges require explicit customer approval. The first beta supports enquiries and quotes; live collection and payouts arrive in the controlled payment phase.

Retail journey

Browse a virtual shop or quick search; open a product card showing real photos, NGN price, stock status and pickup location; request a reservation; merchant confirms availability and expiry; customer collects. Later, enable checkout and merchant fulfilled delivery with explicit fee and coverage. A 3D model is decorative; the product card is authoritative.

Merchant and worker journeys

A business owner chooses retail, artisan or office, uploads a catalogue, sets real coverage, selects a template and submits the profile for review. A worker applies to a funded digital job, receives an assignment, submits files or links, addresses a revision request and gets a verified review and payout eligibility. A merchant may also play using the same account, with separate business permissions.

Daily game loop

Visit the map, check a shop, choose inventory and prices, complete three rotating quests, earn City Coins, visit friends or a club and view weekly standings. NPC demand is server calculated in bounded time steps. Login is never a guarantee of cash income; sponsored missions appear only when a funded campaign has remaining budget.

Scope and release gates

The full requested feature set is retained below. A solo founder should ship in increments rather than treat every feature as a launch dependency.

Stage	Scope	Exit gate

Stage	Scope	Exit gate
A Local vertical slice	Browser map, avatar and profile, three locations, daily quests, City Coins, simulated shop; mock merchant catalogue.	Playable journey works after relo rewards and inventory are valida the server.
B Homelab private beta	Free business spaces, premium template preview and entitlements, catalogues, search, pickup requests, artisan quotes, friends, clubs, leaderboards, real job applications and reviews; admin console.	Invited users complete the three journeys; cross-account access is blocked; backup restore succeed
C Controlled money pilot	Payment partner sandbox then approved live flow, subscriptions, paid orders, job funding, ledger, Nigerian bank payouts, disputes and pre-funded sponsored missions.	Reconciliation, duplicate webhoo payout retry and refund tests pa partner accepts the proposed mo flow.
D Shared city expansion	Real-time multiplayer movement, low-poly 3D buildings and limited interiors.	Target devices and room load tes without affecting commerce accu
E Advanced trading	Complex player-to-player trading of eligible virtual inventory, offer expiry and atomic settlement.	Concurrent trade, disconnect an duplicate acceptance tests pass.

Explicit exclusions from the first release

No cash conversion of ordinary City Coins, speculative land ownership, crypto tokens, wagering, cash player-to-player transfers, delivery fleet, full open world, arbitrary uploaded 3D assets or promises of employment. Premium space is a subscription entitlement, not investment property. Cash earning eligibility is restricted to verified adults in the pilot.

Free and premium spaces

Use one space engine for shops, workshops and offices. Launch three free templates and preview two premium templates: showroom and headquarters. All plans retain basic discovery, enquiries and transaction reviews. Premium can expand catalogue limits, branding and staff access. Prices and limits are configuration, initially tested with merchants; the earlier proposed NGN 5,000 and NGN 15,000 monthly tiers are unvalidated hypotheses.

Timeline approach

Plan milestones rather than guarantee 12 weeks. A suggested part-time founder sequence is 2 to 3 weeks for A, 4 to 6 for B and 4 to 8 or more for C, depending on external approval and testing. D and E require new estimates after profiling. Codex can assist implementation, but the founder reviews money flows, access controls and deployment changes.

Functional requirements and acceptance

ID	Requirement	Acceptance evidence
FR01	Browser map and profiles	Mobile tap and desktop input navigate three hotspots; ava nickname persist; invalid or reported names can be moder

ID	Requirement	Acceptance evidence
FR02	Daily quests and simulated shop	Three quests rotate at a configured daily boundary; one re quest per day; inventory and prices persist; client changes coins.
FR03	Friends clubs leaderboards	Accept or reject friend requests, remove or block users, jo one club; server computes weekly gameplay standings ind real earnings.
FR04	Business spaces and catalogues	Owner edits only their business; shop supports photos, pr stock; service supports coverage and quote request; unsul reverts entitlements without erasing records.
FR05	Discovery and physical commerce	Search by category and service area works outside the gar accepts or rejects reservation; checkout revalidates price a failed payment releases held inventory.
FR06	Jobs submissions and reviews	Only funded live jobs can be assigned in the money phase worker per pilot job; versioned submissions and revision r transaction participants review completed work.
FR07	Wallet and payout	Pending, available and payout-in-progress amounts are vis separately; duplicate requests cannot spend the same bal provider status drives settlement.
FR08	Sponsored missions	Campaign exposes fixed cash reward and eligibility; funde are reserved atomically; exhausted campaigns stop award
FR09	Moderation and disputes	Report user, listing or transaction; admin can hide content eligibility and record decisions; disputed funds cannot be v
FR10	Multiplayer and 3D	Server validates movement; room reconnect works; low-q modes remain usable; interiors load only when entered.
FR11	Advanced virtual trading	Both parties confirm a final offer; items and coins settle to at all; expired, blocked and repeated offers fail safely.

Business records and permissions

Roles: player, business owner, business staff and administrator. Staff permissions are scoped to one business and granted by its owner. Administrator actions require reason codes and immutable audit entries. Paid plan status never marks a business as verified or changes professional review scores.

Notifications and content

Use in-app notifications for applications, quotes, submissions, reviews, disputes and payouts. Start with controlled listing text and moderated uploads. Public chat and voice are outside the first release; friends and clubs do not require unrestricted messaging.

Money economy and transaction design

Three clearly labelled balances

City Coins fund simulation inventory and cosmetic progression and cannot be withdrawn, transferred to cash balances or accepted as payment for real goods. NGN earnings come from completed real transactions and funded sponsored gameplay. XP and reputation record achievement and quality and are neither spendable nor cash convertible.

Cash earning from gameplay

Sponsors buy capped campaigns, such as a merchant product discovery challenge or a verified usability mission. Cleared campaign funds must cover all promised cash rewards before publication. Completion is validated server-side and may enter a review hold. No purchase is required to earn; ordinary quests continue to award City Coins when no sponsors are available. Do not pay rewards from future subscriptions or future user deposits.

Illustrative campaign: NGN 100,000 sponsor payment allocates NGN 60,000 to 600 approved rewards of NGN 100, NGN 20,000 to platform gross income and NGN 20,000 to processing, review and risk provision. This is a configurable planning example, not a provider fee quote or guaranteed margin. If actual costs exceed provision, reduce new campaign capacity rather than change rewards already reserved.

Wallet implementation

Treat the wallet as an earnings ledger with partner payouts, not a general-purpose deposit account. No user top-ups or peer cash transfers in the pilot. Store NGN amounts in integer kobo with currency and immutable double-entry records. Separate pending earnings, available earnings, dispute holds, payout reservations, campaign commitments, platform fees and provider clearing accounts. Each posting has a unique business reference and balanced debits and credits. Corrections use reversal entries.

Payment and payout rules

Use a payment adapter so provider references and statuses are isolated from domain logic. Hosted checkout collects payment data. Verify a payment on the server, match amount, currency and reference, and only then fulfil an order. Validate webhook signatures; durably record each event before acknowledging it; process asynchronously and deduplicate. Initiated transfers remain pending until confirmed; a timeout is an unknown result requiring verification, not a reason to blindly pay again.

For withdrawals, verify the bank recipient and earning eligibility, show fees and minimum amount before confirmation, atomically reserve available earnings and create a stable payout reference. Successful transfer consumes the reservation; confirmed failure releases it; reversal creates an audited compensating entry. Bank account changes trigger re-verification and a risk hold. Never store full identity documents when a provider verification reference will suffice.

Funding and sustainability

Revenue hypotheses: optional premium spaces, transparent commissions on completed orders or bookings, and separately labelled sponsor campaigns. Track contribution per transaction as platform fees less processing, refunds, fraud loss, support and campaign rewards. Customer money, sponsor commitments and earned payouts must be reconciled separately from operating income. Automated rewards pause if campaign liabilities exceed reconciled funds.

Payment launch dependency

Paystack is a candidate adapter, not a confirmed partner. Its documentation describes Nigerian bank recipients, server payment verification and transfer status webhooks. Marketplace settlement, holding funds, identity requirements and sponsored reward payouts require explicit partner confirmation and Nigeria-specific legal review before live mode. Do not advertise escrow or payment protection until the contracted flow actually supports it.

Operations fraud and state transitions

Transaction states

- Job: draft → awaiting funding → open → assigned → submitted → revision requested or accepted → earnings pending → available. A dispute can pause settlement before payout.
- Retail order: reserved → awaiting payment → paid → ready for pickup or dispatched → completed. Cancellation and refund are separate audited transitions, including partial refunds where supported.
- Service booking: request → quoted → accepted → scheduled → completion submitted → customer confirmed → settlement eligible. Approved quote revisions are retained as versions.
- Reward claim: eligible → budget reserved → evidence submitted → approved or rejected → pending earnings → available. Unused reservations expire and release budget.

Fraud controls

Enforce one verified earning identity, limits per identity and campaign, task cooldowns and server timestamps. Use device, account and network risk signals to route reviews, not automatically penalise shared Nigerian mobile networks. Detect duplicate submissions, repeated recipient accounts, fake fulfilment and merchant-worker collusion. Randomise task evidence where practical. Newly verified earners receive lower limits and manual payout review.

Disputes and moderation

A participant selects a reason, attaches evidence and sees a proposed response deadline, initially 48 hours. Admin records evidence, responses, decision and money movement. Freeze the disputed amount while preserving unrelated earnings. Proposed internal target: initial review within two business days; validate founder capacity before promising this publicly. Appeals have an independent second review where feasible, even if supplied by a contracted operator.

Provide report and block controls; suspend abusive accounts; hide unsafe or fraudulent listings. Business verification communicates exactly what was checked. Trades, high game levels and premium subscriptions do not establish artisan qualifications. Do not publish a worker home address on their public profile.

Security and recovery

Require secure sessions, role checks on every object, admin MFA, rate limiting, TLS, signed upload URLs, allowed upload types, malware scanning for work files and secrets outside source control. Record audit history for bank changes, payouts, moderation and plan changes. Exclude secrets and identity evidence from application logs. Retention and deletion settings must be reviewed before live launch.

Required money tests

Replay the same webhook ten times; credit once. Request two withdrawals against one balance concurrently; only valid funds reserve. Simulate a transfer timeout then eventual success; no second payment. Test reversal after success, refund after earnings allocation, campaign exhaustion and concurrent stock purchase. Restart the worker

during processing and verify safe recovery. Reconcile provider statement to ledger daily; unresolved differences block affected payouts.

Release switches

Separate switches control checkout, withdrawals, sponsored cash rewards, multiplayer and trading. A payment incident pauses new affected transactions while preserving read-only history and support access. Game coins must remain usable even when the cash subsystem is paused.

Architecture and homelab deployment

Proposed stack: TypeScript web application and modular API, PostgreSQL, background worker, optional Redis for queues and presence, and object storage for images and submissions. Use a lightweight 2D map first; add a Three.js client for 3D later. Browser rendering uses the player device GPU, so the server needs no GPU. Verify package versions and integration details during implementation.

System boundaries

Web and game clients call one authorised API. API modules cover identity, businesses, catalogue, discovery, orders, bookings, jobs, game economy, social, ledger and admin. A worker handles notifications, webhook processing, reconciliation and payout jobs. PostgreSQL is the source of truth for balances, inventory and transactions. Redis presence or caches may be rebuilt; they never own cash balances.

Data model

Core entities: users, profiles, avatars, businesses, business_members, space_templates, subscriptions, products, services, inventory_reservations, orders, order_items, service_requests, quote_versions, bookings, jobs, applications, assignments, submissions, reviews, friendships, clubs, memberships, quest_definitions, quest_completions, game_ledger_entries, campaigns, reward_claims, journal_entries, journal_lines, payout_recipients, payouts, webhook_events, disputes and audit_events. Later add rooms, presence sessions and trade offers.

Development and deployment sizing

Environment	Planning allocation	Purpose
Development PC	Prefer 16 GB total RAM and 30 to 50 GB free SSD; smaller setups can run API and database while disabling optional services.	Local editor, browser and containers. Actual PC specifications need confirmation.
Dell private beta VM	Start 4 vCPU, 6 GB RAM, 60 GB SSD; increase to 8 GB RAM if other VM allocations permit.	One Ubuntu VM with Docker Compose API, web, worker and PostgreSQL; invite-only testing.
Lenovo support node	Use existing services; add only small monitoring or backup tasks if capacity permits.	Avoid placing the primary database here until RAM and storage upgrades are confirmed.

These are starting allocations, not measured capacity guarantees. Your Dell was previously described as an i5-10500T with 16 GB RAM and 256 GB disk. Check current free memory, storage and existing VMs before allocating. A

VM is the recommended first deployment; a Proxmox cluster, Kubernetes and shared storage are unnecessary for this phase.

PC to homelab path

- Keep a single Git repository and separate local, staging and future production configuration. Containerise services and provide one Compose definition with health checks and persistent volumes.
- Build and test locally; push an immutable image version; deploy into the Dell VM; back up PostgreSQL and run migrations; check login, map, job submission and payment sandbox; record the deployed version.
- For initial access use LAN or private VPN. Payment sandbox webhooks need a reachable HTTPS endpoint or supported development relay. Never expose Proxmox, database ports or Docker control ports publicly.
- Back up database and uploads off the Dell; encrypt backups; perform a restore on a separate test database. VM snapshots supplement backups but do not replace them. Roll back application images only with schema compatibility checked.

Backlog validation and build sequence

Order	Deliverable	Done when
1	Repository and platform foundation	Local startup, schema migrations, seed data, sign-in, ro checks and CI build work.
2	Playable city slice	Map, avatar, three hotspots, one simulated shop and th quests work with server state.
3	Merchant space engine	Three free templates, two premium previews, catalogu owner dashboard and moderation work.
4	Commerce and work journeys	Pickup reservation, artisan quote, job application, subm verified review complete end to end.
5	Social features	Friend controls, clubs, blocking and weekly server comp work.
6	Homelab beta operations	Versioned images deploy; restricted access, metrics, off and restoration are demonstrated.
7	Payment sandbox and ledger	Funding, callbacks, webhooks, refunds, payouts and co pass with exact reconciliation.
8	Live money pilot	Partner flow and eligibility approved; limited users, pay funded sponsor missions and disputes operational.
9	Multiplayer and 3D	Small room movement and limited interiors pass mobil room load tests.
10	Complex virtual trading	Atomic offers, cancellation, expiry, item locks and fraud

Performance and reliability targets

Proposed test targets: marketplace page useful within 3 seconds on a throttled mid-range Android test device; non-payment API p95 below 500 ms for 25 concurrent active users; initial map assets below 5 MB compressed. Multiplayer begins at 10 concurrent players per room, then tests 20 before increasing. These targets need measurements and may change. External payment latency is reported separately from API performance.

Private beta recovery goals: daily backup gives up to 24 hours of data loss and a restore target of 4 hours, tested rather than promised. Before live money use, shorten the recovery point, retain durable provider events and verify transaction recovery procedures. Track CPU, memory, disk, queue age, API errors and reconciliation gaps.

Analytics and pilot decisions

Record onboarding completion, first shop action, quest completion, friend acceptance, catalogue view, enquiry, reservation, quote acceptance, job completion, funded reward and payout result. Avoid sending financial or identity details to general analytics. Compare users who play without earning with earning users to test whether entertainment stands independently.

Founder workflow with Codex

Implement one backlog row at a time. Give Codex the relevant requirements and ask for an implementation plan, code, migrations and acceptance checks. Review changes in Git and exercise the user journey before merging. Use seeded businesses and fake balances locally; keep live credentials out of coding prompts. The first task should be the map plus avatar and simulated shop, not the entire money platform.

Open decisions before payment development

Confirm operating entity and provider acceptance; merchant service area and onboarding checks; cash reward sponsor commitments; commission and subscription experiments; completion and refund policy; verified adult requirements; public hosting and backup budget. Business membership terms and applicable privacy, consumer and payment obligations need appropriate review.

Technical source references

Paystack documentation checked 11 October 2026. These references support integration mechanics only; they do not establish approval of the Hustle City business model.

https://paystack.com/docs/payments/verify-payments/

https://paystack.com/docs/payments/webhooks/

https://paystack.com/docs/transfers/single-transfers/

https://paystack.com/docs/transfers/bulk-transfers/